

PART 1: AI CONCEPTS REPORT

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. AI systems can learn from data, understand language, recognize images, solve problems, and make decisions. Today AI is used in virtual assistants, recommendation systems, healthcare, education, banking, and many other fields. The main goal of AI is to make machines capable of performing intelligent tasks efficiently.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

Feature	Artificial Intelligence(AI)	Machine Learning(ML)	Generative AI (Gen AI)	Agentic(AI)
Definition	A broad field of computer science that enables machines to perform tasks that normally require human intelligence.	A subset of AI that allows systems to learn from data and improve performance without explicit programming	A type of AI that generates new content such as text, images, videos, audio, and code.	An advanced AI system that can make decisions, plan actions, and execute tasks autonomously.
Main Goal	Simulate human intelligence	Learn patterns from data and make predictions	Create original content	Complete tasks and achieve objectives independently
Data requirement	May require data depending on the application	Requires training data	Requires very large datasets for training	Uses training data ,memory ,and real-time information

Q3. Explain any three open-source AI models available on Hugging Face.

1. Llama: Llama is an open-source language model developed by Meta. It can generate text, answer questions, summarize content, and assist in various AI applications.

2. Mistral: Mistral is a lightweight and efficient open-source model known for strong performance while using fewer computing resources. It is widely used in chatbots and AI assistants.

3. BLOOM: BLOOM is a multilingual language model that supports many languages. It can generate text, translate content, and help researchers build AI applications.

Q4. Difference between traditional programming and AI-assisted development

Traditional programming relies on developers writing explicit instructions and rules. The computer follows those rules exactly to produce results. If requirements change, developers must manually update the code. In AI-assisted development, AI tools help developers write code, identify bugs, generate suggestions, and automate repetitive tasks. This improves productivity and reduces development time. However, developers still need to verify and review AI-generated results for accuracy.

Q5. Explain AI hallucinations with one example.

AI hallucination occurs when an AI system generates information that appears correct but is actually false, misleading, or unsupported by facts. Hallucinations happen because AI predicts likely responses based on patterns in data instead of truly understanding information.

Example: A student asks an AI tool for references on a topic. The AI may generate a book title, author name, and publication details that do not actually exist. If the student uses this information without verification, it can lead to errors. Therefore, important AI-generated information should always be checked against trusted sources.